

Introduction to Inference, Examples

CSU, Chico Math 314

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outline

Confidence Interval

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Confidence Interval, example

Consider a data set consisting of number of users connected to a server on a randomly chosen day. Measurements were taken every minute for 100 minutes starting at a randomly chosen time. The data indicated a sample mean of $\bar{X} = 137$ users with a population standard deviation of 40. Create a 95% confidence interval for the mean number of users connected to this server.

General Tip: Identify all your relevant variables and define them in R to help with your setup for the problem.

Hypothesis Test, example

The company that owns this server claims that on any given day 150 users are connected to its server, but you believe that claim is **not** true¹. Test this claim at the significance level of $\alpha = 0.05$.

- ▶ State hypotheses and choose level of significance.
- ▶ Collect data.
- ▶ Calculate summary statistics and p-value.
- ▶ Conclude.

¹A more realistic hypothesis test for the company might attempt to claim that more than 150 customers connect on any given day – for marketing purposes.

Confidence Intervals related to Hypothesis Tests

When α **matches** between the confidence interval and hypothesis test, and the hypothesis test has a **two-sided alternative**, we can safely compare the confidence interval to the hypothesis test.

- ▶ Did the confidence interval contain the null value?
- ▶ Which hypothesis then seems more likely given the confidence interval?

Confidence Intervals related to Hypothesis Tests, be careful

This relationship between confidence intervals and hypothesis tests is tricky when the alternative hypothesis is **one-sided** ($H_1 : \mu < \mu_0$ or $H_1 : \mu > \mu_0$). If you have a two-sided confidence interval, it will not accurately represent a one-sided hypothesis test. However, you can still use a **one-sided** confidence interval for these cases.

One-Sided Confidence Intervals

One-sided confidence intervals will have a different construction depending on the equivalent hypothesis test:

If testing hypothesis tests with the alternative hypothesis $H_1 : \mu > \mu_0$, use an **upper** confidence interval:²

$$[\bar{X} - z_{1-\alpha}^* \cdot \sigma_{\bar{X}}, \infty].$$

If testing hypothesis tests with the alternative hypothesis $H_1 : \mu < \mu_0$, use a **lower** confidence interval:

$$[-\infty, \bar{X} + z_{1-\alpha}^* \cdot \sigma_{\bar{X}}].$$

Note that these confidence intervals will extend to infinity in one direction, yet they will still contain the same area under the curve expected of a $100(1-\alpha)\%$ confidence interval.

²As seen in the main slides, $z_{1-\alpha}^* = qnorm(1 - \alpha)$.

Obtaining the p-value for One-Sided Tests

For the two-sided case, the p-value was found as follows:

```
2*(1 - pnorm(abs(z))) # 2-sided cases
```

For one-sided cases, if testing hypothesis tests with the alternative hypothesis $H_1 : \mu > \mu_0$:

```
1 - pnorm(z) # 1-sided case
```

And if testing hypothesis tests with the alternative hypothesis $H_1 : \mu < \mu_0$:

```
pnorm(z) # 1-sided case
```


Hypothesis Test, example revisited

Let's revisit the example from slide 4 with the more realistic hypothesis in play. The company that owns this server wants to claim that on any given day **more than** 150 users are connected to its server. Test this claim at the significance level of $\alpha = 0.05$.

- ▶ State hypotheses and choose level of significance.
- ▶ Collect data.
- ▶ Calculate summary statistics and p-value.
- ▶ Conclude.

One-Sided Confidence Intervals, Bernoulli Case

When working with **proportions**, the confidence intervals won't extend to ∞ , since p can only take values between 0 and 1. Instead, one-sided confidence intervals will be bound by these limitations in place of infinity.

If testing hypothesis tests with the alternative hypothesis $H_1 : p > p_0$, use an **upper** confidence interval:

$$[\hat{p} - z_{1-\alpha}^* \cdot \sigma_{\bar{X}}, 1].$$

If testing hypothesis tests with the alternative hypothesis $H_1 : p < p_0$, use a **lower** confidence interval:

$$[0, \hat{p} + z_{1-\alpha}^* \cdot \sigma_{\bar{X}}].$$

Why Confidence Intervals?

The purpose of a confidence interval is to give a more informative estimator than a simple point estimator. You can achieve a certain level of confidence that your interval contains the true population mean, rather than hoping that your point estimator happens to be fairly close.

Remember, there are two parts to the margin of error in a two-sided confidence interval: **number of standard deviations** $z_{1-\alpha/2}^*$, and **size of the standard deviations** $\sigma_{\bar{X}}$.

Why Hypothesis Tests?

When you want to test a statement (H_0), a hypothesis test lets you compare your sample to the statement by determining how unlikely your test statistic is to occur under the assumption that statement is true.

We find the z-score for our test statistic $\bar{X}$ or $\hat{p}$, which represents how many standard deviations away from the mean the test statistic is.

By using `pnorm()` to find how unlikely such a test statistic is to occur, in relation to some level of confidence α , we can determine whether we reject or fail to reject the statement in question.

references I

Adapted from notes by Dr. Roualdes.